

Noise Hotspot Identification and Deeper Analysis in Kolkata

B.Stat–III Statistics Comprehensive Group Project

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Sound and Sound Levels

Sound

- Sound is a pressure wave propagating through a medium such as air.
- Environmental sound levels are commonly measured in decibels (dB).

Sound Pressure Level (SPL)

- Quantifies sound intensity relative to a reference pressure.

The sound pressure level is defined as:

$$L_p = 20 \log_{10} \left(\frac{p}{p_0} \right)$$

where:

- p = measured sound pressure,
- $p_0 = 20 \mu\text{Pa}$ (reference sound pressure in air).

Interpretation

- Higher dB values correspond to louder sounds.
- The decibel scale is logarithmic.

Instantaneous Sound Levels

Instantaneous Sound Level

- Represents the sound pressure level measured at a particular instant of time.
- Environmental noise fluctuates continuously due to:
 - traffic,
 - loudspeakers,
 - firecrackers,
 - crowd activity,
 - industrial sources.

Characteristics

- Highly time-varying.
- May contain sharp spikes and sudden drops.
- Difficult to interpret directly over long durations.

Hence, equivalent continuous sound measures are often preferred for environmental analysis.

Equivalent Continuous Sound Level (Leq)

Equivalent Continuous Sound Level (Leq)

- Represents the constant sound level that contains the same acoustic energy as the varying sound over a specified time interval.

Mathematically,

$$L_{eq} = 10 \log_{10} \left(\frac{1}{T} \int_0^T 10^{L(t)/10} dt \right)$$

where:

- $L(t)$ = instantaneous sound level at time t ,
- T = total observation duration.

Importance

- Widely used in environmental noise monitoring.
- Provides a single summary measure over a time period.

Frequency Weighting in Noise Measurement

Human hearing does not respond equally to all frequencies.

To account for this, sound measurements often use **frequency weighting filters**.

A-weighting

- Approximates human hearing sensitivity.
- Reduces emphasis on very low and very high frequencies.
- Commonly used in environmental noise regulations.

C-weighting

- Applies much less frequency attenuation.
- Better captures low-frequency and impulsive sounds.

Z-weighting

- Zero-frequency weighting.
- Essentially an unweighted measurement.

LAeq, LCeq and LZeq

LAeq

- Equivalent continuous sound level using A-weighting.
- Most commonly used environmental noise metric.

LCeq

- Equivalent continuous sound level using C-weighting.
- More sensitive to low-frequency and impulsive sounds.

LZeq

- Equivalent continuous sound level using Z-weighting.
- Represents essentially unweighted sound energy.

Interpretation in Urban Noise Context

- LAeq:
 - traffic,
 - crowd noise,
 - general environmental exposure.

Other Common Noise Metrics

- **LCeq:**
 - loud music,
 - bass-heavy amplification,
 - firecracker activity.
- **LZeq:**
 - raw physical sound energy assessment.

LApeak

- Peak instantaneous LAeq sound pressure level.
- Useful for impulsive noise such as firecrackers.

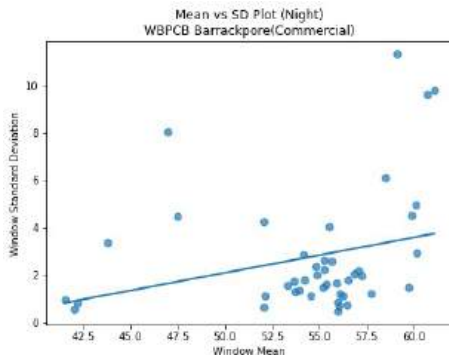
Checking if Additive Model is Fine

In order to check if additive model is appropriate in our case, we check the Mean v/s Standard Deviation plots for 2 noise monitoring stations.

As we would see in the following plots, the Mean-SD plots showed no strong proportional increase in variability with increasing mean levels. Although a weak positive association was observed in some cases, the variance remained approximately stable across different mean values. Hence, the time series was considered more consistent with an additive structure rather than a multiplicative one.

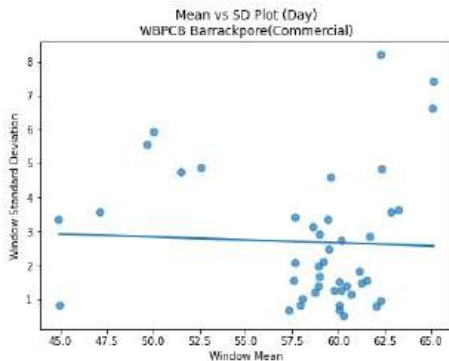
WBPCB Barrackpore – Commercial Zone (Night)

- The nighttime commercial data shows somewhat higher variability at larger mean levels compared to daytime observations.
- Although a mild positive relationship is visible, the increase in variance is not excessively strong.
- Therefore, the additive decomposition model can still be considered acceptable for practical analysis.



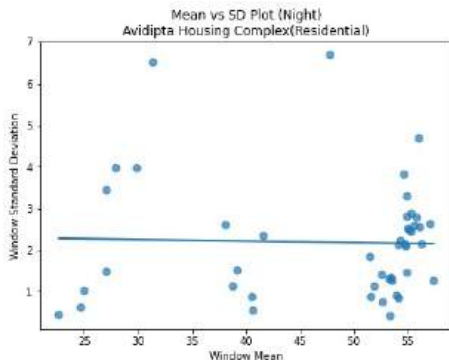
WBPCB Barrackpore – Commercial Zone (Day)

- The scatter plot shows a relatively stable spread of standard deviation across varying mean levels.
- The fitted trend line is nearly flat with a slight negative slope, indicating weak dependence between mean and variability.
- This behavior supports the suitability of an additive decomposition model for this series.



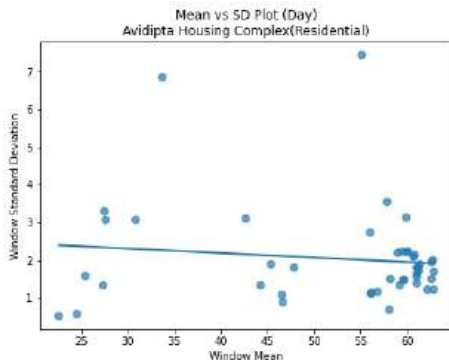
Avidipta Housing – Residential Zone (Night)

- The nighttime residential data exhibits moderate dispersion with no strong increasing variance pattern.
- Most observations are clustered around similar standard deviation levels despite changes in the rolling mean.
- The nearly horizontal fitted line suggests approximate variance stability, supporting the additive assumption.



Avidipta Housing – Residential Zone (Day)

- Daytime residential observations show slight variability in standard deviation across the range of means.
- However, the absence of a strong upward trend indicates that variance does not systematically increase with the mean.
- This suggests that the additive model remains reasonably appropriate for daytime residential noise data.



Stationarity and ADF Test

- A time series is said to be **stationary** if its statistical properties remain constant over time.
- In a stationary series:
 - Mean remains constant
 - Variance remains stable
 - Autocovariance depends only on lag
- The **Augmented Dickey-Fuller (ADF) Test** is used to check for the presence of a unit root.
- Hypotheses of the ADF test:
 - H_0 : Series is non-stationary
 - H_1 : Series is stationary
- If the p-value is less than 0.05, the null hypothesis is rejected and the series is considered stationary.

ADF Test Results – Daytime Noise Data

Station	ADF Stat.	p-value	Conclusion
Avidipta Housing	0.6643	0.9891	Non-Stationary
Ballygunge CU Campus	-3.0224	0.0329	Stationary
Bandel Thermal WBPDC	-1.7865	0.3872	Non-Stationary
WBPCB Barrackpore	-1.6528	0.4556	Non-Stationary
Howrah Municipal	-2.5904	0.0950	Non-Stationary
Jalan Complex	-1.7740	0.3934	Non-Stationary
Leather Complex	-3.6370	0.0051	Stationary
Loreto College	-4.6218	0.0001	Stationary
Presidency University	-3.1489	0.0231	Stationary
Shyam Power LTD	-3.6963	0.0042	Stationary
Solaris Ideal Riverview	-2.1859	0.2114	Non-Stationary
Urbana Housing	-4.0074	0.0014	Stationary

ADF Test Results – Nighttime Noise Data

Station	ADF Stat.	p-value	Conclusion
Avidipta Housing	0.1378	0.9685	Non-Stationary
Ballygunge CU Campus	-2.9546	0.0394	Stationary
Bandel Thermal WBPCL	-1.9093	0.3277	Non-Stationary
WBPCB Barrackpore	-1.8669	0.3478	Non-Stationary
Howrah Municipal	-2.4283	0.1339	Non-Stationary
Jalan Complex	-1.7563	0.4024	Non-Stationary
Leather Complex	-3.3374	0.0133	Stationary
Loreto College	-4.2094	0.0006	Stationary
Presidency University	-3.1484	0.0232	Stationary
Shyam Power LTD	-5.2125	0.0000	Stationary
Solaris Ideal Riverview	-3.0506	0.0304	Stationary
Urbana Housing	-5.3209	0.0000	Stationary

Interpretation of ADF Test Results

- The ADF test results indicate that both stationary and non-stationary behavior are present across different monitoring stations.
- Several institutional and industrial stations such as:
 - Loreto College
 - Presidency University
 - Leather Complex
 - Shyam Power LTD

show significant evidence of stationarity with p-values below 0.05.

- In contrast, a number of residential and commercial stations exhibit non-stationary characteristics, suggesting the presence of:
 - Long-term trends
 - Seasonal effects
- Nighttime data appears relatively more stationary compared to daytime observations, likely due to reduced traffic and human activity during night hours.

Seasonal Analysis

- In this study, environmental noise data has been collected from the official monitoring network of the West Bengal Pollution Control Board (WBPCB).
- The analysis focuses on seasonal and temporal variations in noise levels using data from selected monitoring stations across different acoustic zones.
- The study aims to identify recurring seasonal patterns and understand long-term noise dynamics through statistical time series analysis.

Selected Monitoring Stations

Residential Zones

- Avidipta Housing
- Solaris Ideal Riverview
- Urbana Housing

Commercial Zones

- Leather Complex
- Howrah Municipality
- WBPCB Barrackpore

Industrial Zones

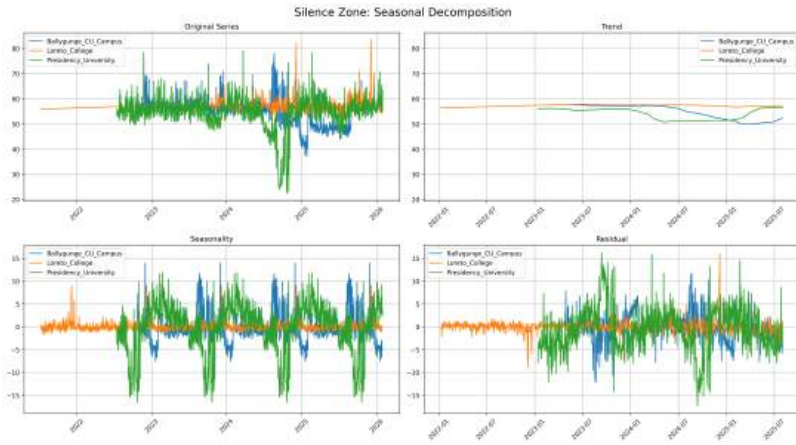
- Bandel Thermal WBPDCCL
- Jalan Complex
- Shyam Power LTD

Institutional / Silence Zones

- Ballygunge CU Campus
- Loreto College
- Presidency University

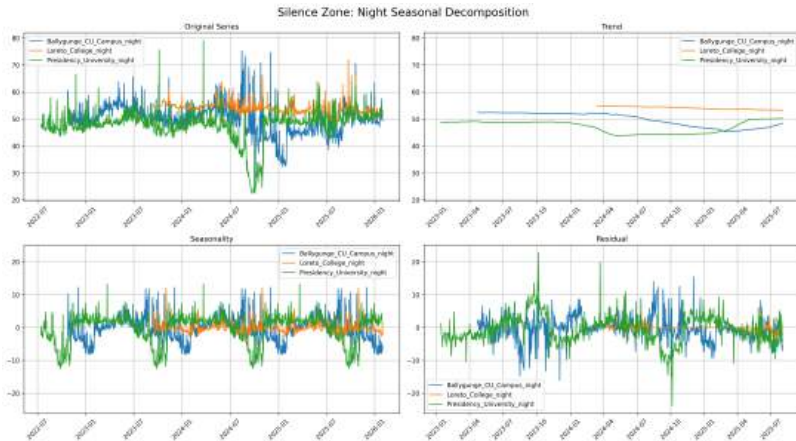
Silence Zone – Daytime Noise Pattern

- The following plot shows the seasonal and temporal variation of daytime noise levels in Silence Zone monitoring stations.
- Silence zones generally include educational and institutional regions where stricter noise regulations are expected.



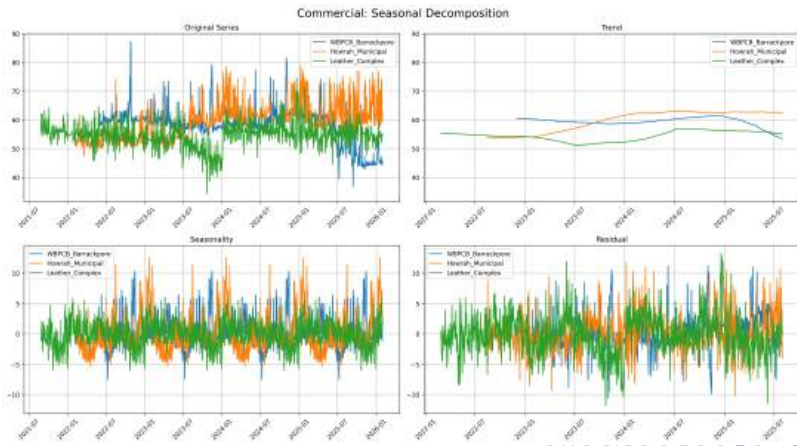
Silence Zone – Nighttime Noise Pattern

- This figure illustrates nighttime noise behavior in Silence Zone stations.
- Reduced human activity during nighttime is expected to produce lower variability and comparatively lower noise levels.



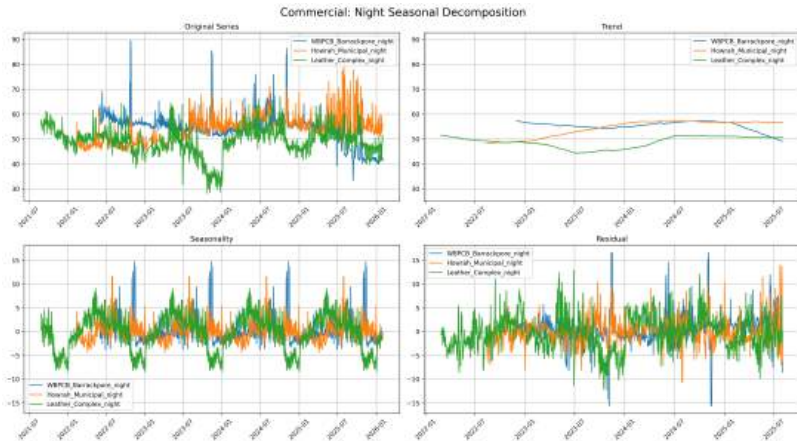
Commercial Zone – Daytime Noise Pattern

- Commercial regions generally experience elevated daytime noise due to: Traffic congestion, market activity and public movement.
- The plot highlights seasonal fluctuations in commercial daytime noise levels.



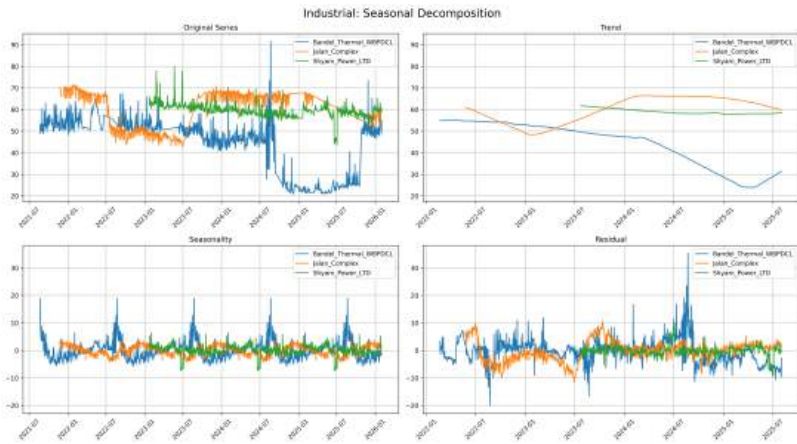
Commercial Zone – Nighttime Noise Pattern

- The nighttime commercial noise profile reflects reduced economic activity and traffic intensity.
- Residual fluctuations may still persist due to urban transport and late-night commercial operations.



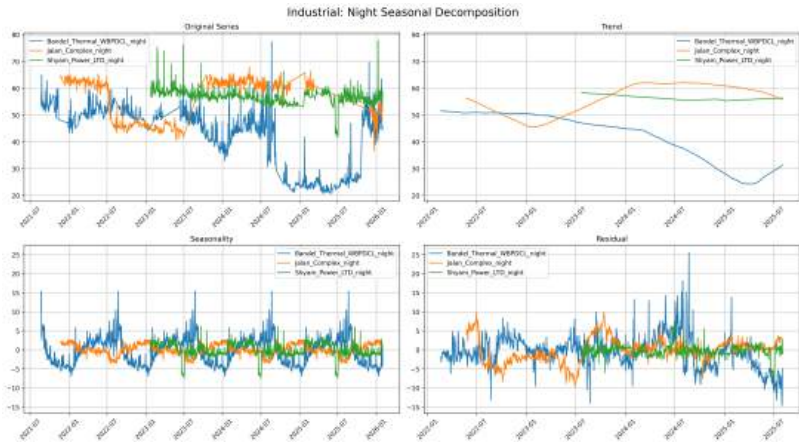
Industrial Zone – Daytime Noise Pattern

- Industrial regions typically show high daytime noise levels because of: Machinery operation, heavy transport and manufacturing activity.
- The figure captures seasonal trends and operational variability in industrial noise.



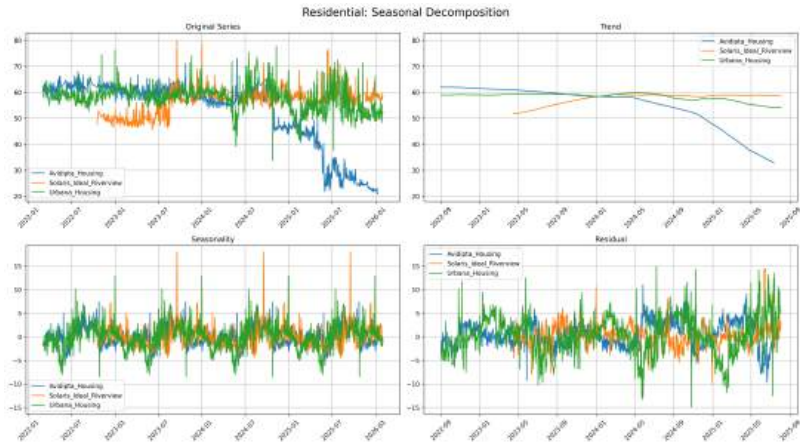
Industrial Zone – Nighttime Noise Pattern

- This plot illustrates nighttime industrial noise characteristics.
- Continuous industrial operations and freight movement may contribute to sustained nighttime noise levels.



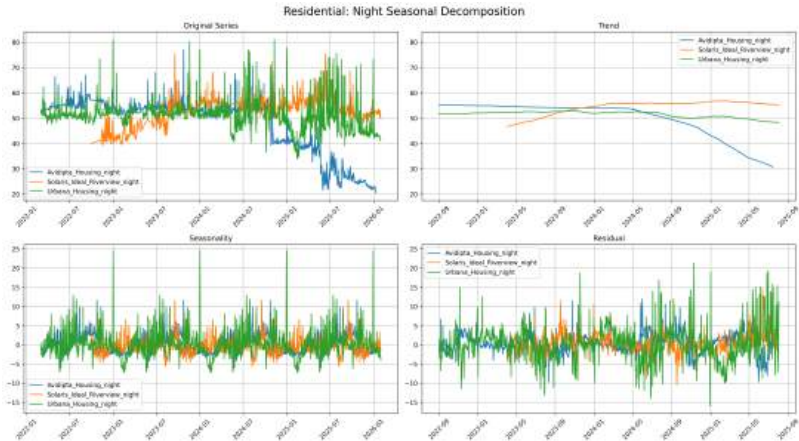
Residential Zone – Daytime Noise Pattern

- Residential zones exhibit moderate daytime noise influenced by local transportation, human activity, construction and neighborhood movement.
- The plot demonstrates temporal and seasonal variability in residential areas.



Residential Zone – Nighttime Noise Pattern

- Nighttime residential noise levels are expected to decline due to reduced outdoor activity.
- The seasonal structure helps identify persistent urban noise exposure during night hours.



Identification of Seasonal/Monthly Hotspots Across the Greater Kolkata Region Based on Data from 2023–2025

Introduction

Noise Hotspot:

- A locality where monitoring stations consistently exhibit elevated noise levels relative to prescribed standards.
- Indicates persistent or recurring environmental noise stress.

Noise Coldspot:

- A locality where monitoring stations consistently exhibit comparatively lower noise levels.
- Indicates relatively quieter environmental conditions.

Objective:

- Identify spatial clusters of high and low exceedance behaviour across the Greater Kolkata region using monitoring station data from 2023–2025.

What Measures Are We Using Here?

For each monitoring station and date:

- Compute the equivalent continuous sound level (**LAeq**) separately for:
 - **Day period:** 6 AM – 10 PM
 - **Night period:** 10 PM – 6 AM
- Compare the computed values with the prescribed noise limits given by the **Central Pollution Control Board (CPCB)** for the corresponding zone.
- For a month or festive season, calculate:

$$\text{Exceedance Proportion} = \frac{\text{Number of days exceeding CPCB limits}}{\text{Total number of observed days}}$$

- Daytime and nighttime exceedance proportions are analysed separately.

Monitoring Stations Accumulating in a Region

- Nearby monitoring stations may collectively display similar exceedance behaviour.
- If several neighbouring stations simultaneously exhibit:
 - high exceedance proportions → potential hotspot
 - low exceedance proportions → potential coldspot
- Such spatial clustering may indicate:
 - persistent environmental noise pressure,
 - localized festival-related amplification,
 - traffic or urban activity concentration.
- These clusters can help identify localities where mitigation and regulatory measures may be necessary.

Festive Periods Considered

Noisy festive periods considered include:

- Holi (loud music dominated)
- Durga Puja / Dussehra period
- Diwali (firecracker dominated)
- Christmas + New Year celebrations

These windows were selected to capture periods associated with increased community noise activity and elevated sound exposure.

Festive Periods: 2023

Festival/Event	Period
Holi	March 3 – March 14
Durga Puja + Dussehra	October 14 – October 31
Diwali Spillover	November 2 – November 17
Christmas + New Year	December 22 – December 31
New Year Carry-over	January 1 – January 5

Festive Periods: 2024

Festival/Event	Period
Holi	March 19 – March 30
Durga Puja + Dussehra	October 2 – October 18
Diwali Spillover	October 21 – November 10
Christmas + New Year	December 20 – December 31
New Year Carry-over	January 1 – January 5

Festive Periods: 2025

Festival/Event	Period
Holi	March 8 – March 19
Durga Puja + Dussehra	September 21 – October 9
Diwali Spillover	October 15 – October 28
Christmas + New Year	December 20 – December 31
New Year Carry-over	January 1 – January 5

Global and Local Spatial Autocorrelation

Spatial Autocorrelation refers to the tendency of geographically nearby observations to exhibit similar values.

Positive Spatial Autocorrelation

- Nearby locations tend to have similar values.
- Example:
 - high exceedance stations clustering together,
 - low exceedance stations clustering together.

Negative Spatial Autocorrelation

- Nearby locations tend to have dissimilar values.

Global Spatial Autocorrelation

- Measures overall spatial dependence across the entire study region.

Local Spatial Autocorrelation

- Identifies localized clusters and spatial anomalies.

Moran's I: Detecting Global Spatial Autocorrelation

We first test whether exceedance proportions exhibit global spatial dependence.

Suppose:

- x_i = exceedance proportion at station i
- $\bar{x}$ = mean exceedance proportion
- w_{ij} = spatial weight between stations i and j
- n = number of stations

The Moran's I statistic is:

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}$$

Moran's I: Hypothesis and Decision Rule

Null Hypothesis (H_0)

- No global spatial autocorrelation exists.
- Observed exceedance proportions are spatially random.

Alternative Hypothesis (H_1)

- Spatial autocorrelation is present.

Interpretation

- $I > 0$: positive spatial autocorrelation
- $I < 0$: negative spatial autocorrelation
- $I \approx 0$: spatial randomness

Decision Rule

- Compute permutation-based p-value.
- Reject H_0 if:

$$p < \alpha$$

where α is the chosen significance level.

Moran's I — Festive Periods

Festival	Year	Period	Moran's I	z	p (sim)	Decision
*Diwali	2023	Day	0.5860	3.653	0.001	Rejected
	2023	Night	0.1748	1.371	0.092	Rejected
	2024	Day	0.4384	3.008	0.005	Rejected
	2024	Night	-0.0069	0.105	0.411	Not rejected
	2025	Day	0.0763	1.123	0.137	Not rejected
	2025	Night	0.0209	0.408	0.319	Not rejected

Table: *

H_0 : No spatial autocorrelation in LAeq exceedance proportions. $\alpha = 0.10$, permutation-based p -value (999 permutations).

Moran's I — Festive Periods

Festival	Year	Period	Moran's I	z	p (sim)	Decision
*Durga Puja	2023	Day	0.6062	3.343	0.003	Rejected
	2023	Night	0.0333	0.513	0.216	Not rejected
	2024	Day	0.5356	3.648	0.001	Rejected
	2024	Night	0.0125	0.258	0.352	Not rejected
	2025	Day	0.0993	1.477	0.073	Rejected
	2025	Night	0.0758	1.202	0.120	Not rejected

Table: *

H_0 : No spatial autocorrelation in LAeq exceedance proportions. $\alpha = 0.10$, permutation-based p -value (999 permutations).

Moran's I — Festive Periods

Festival	Year	Period	Moran's I	z	p (sim)	Decision
Christmas & New Year	2023	Day	0.4019	2.530	0.012	Rejected
	2023	Night	0.2423	1.551	0.080	Rejected
	2024	Day	0.4256	2.774	0.006	Rejected
	2024	Night	-0.0732	-0.346	0.399	Not rejected
	2025	Day	-0.0561	-0.569	0.288	Not rejected
	2025	Night	-0.0461	-0.436	0.353	Not rejected

Table: *

H_0 : No spatial autocorrelation in LAeq exceedance proportions. $\alpha = 0.10$, permutation-based p -value (999 permutations).

Local Spatial Autocorrelation

After verifying the presence of global spatial autocorrelation, we investigate localized clustering behaviour.

- Global statistics provide only an overall summary.
- They do not identify:
 - where clusters occur,
 - which stations contribute strongly to clustering.
- Local spatial autocorrelation statistics allow identification of:
 - hotspot regions,
 - coldspot regions,
 - localized spatial anomalies.

For this purpose, we use the **Getis–Ord G_i^* statistic**.

Getis-Ord G_i^* Statistic

For station i , the Getis-Ord G_i^* statistic is:

$$G_i^* = \frac{\sum_j w_{ij} x_j - \bar{X} \sum_j w_{ij}}{S \sqrt{\frac{n \sum_j w_{ij}^2 - (\sum_j w_{ij})^2}{n-1}}}$$

where:

- x_j = exceedance proportion at station j
- w_{ij} = spatial weight between stations
- $\bar{X}$ = global mean
- S = global standard deviation

The statistic evaluates whether high or low values cluster spatially around a station.

Getis-Ord G_i^* : Hypothesis and Decision Rule

Null Hypothesis (H_0)

- No local spatial clustering exists around station i .

Alternative Hypothesis (H_1)

- Significant local clustering exists around station i .

Interpretation

- Large positive G_i^* :
 - hotspot,
 - clustering of high exceedance values.
- Large negative G_i^* :
 - coldspot,
 - clustering of low exceedance values.

Decision Rule

- Compute corresponding z-scores and p-values.
- Significant positive z-score:
 - hotspot detected.
- Significant negative z-score:
 - coldspot detected.

Getis-Ord-Gi*

Here, we use a significance level of 10% for the Getis-Ord-Gi* tests.
We use a binary spatial weights matrix, with distance radius =4km.

Comparison of Hotspot Maps



Figure: Daytime Hotspots



Figure: Nighttime Hotspots

Comparison of Hotspot Maps

Gi* Hotspots/Coldspots
Christmas NewYear 2024 &C" Day period



Gi* Hotspots/Coldspots
Christmas NewYear 2024 &C" Night period



Comparison of Hotspot Maps

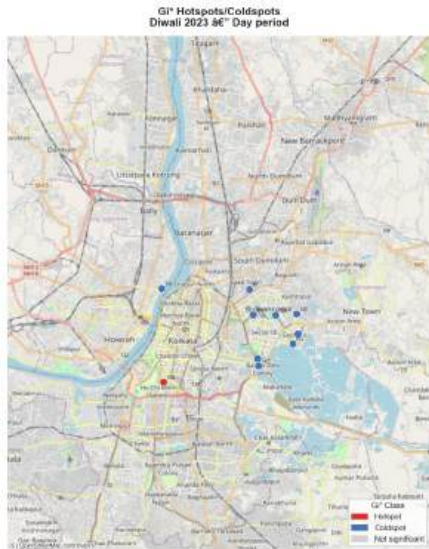


Figure: Daytime Hotspots



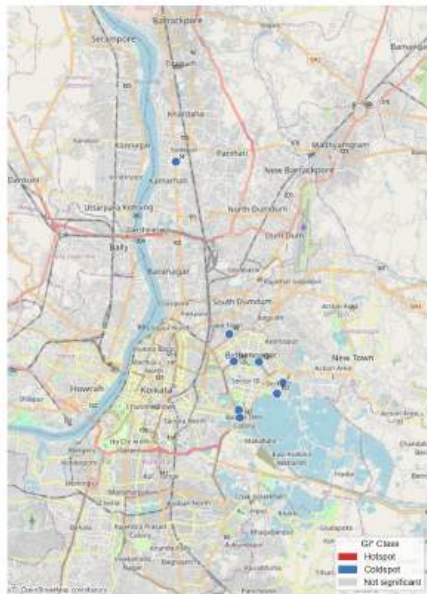
Figure: Nighttime Hotspots

Comparison of Hotspot Maps

GI* Hotspots/Coldspots
Diwali 2024 at Day period

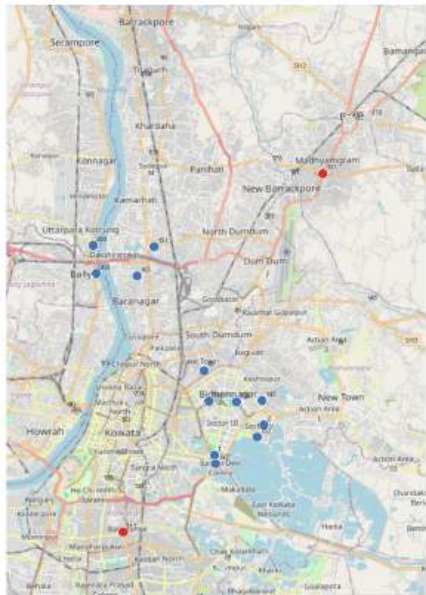


GI* Hotspots/Coldspots
Diwali 2024 at Night period

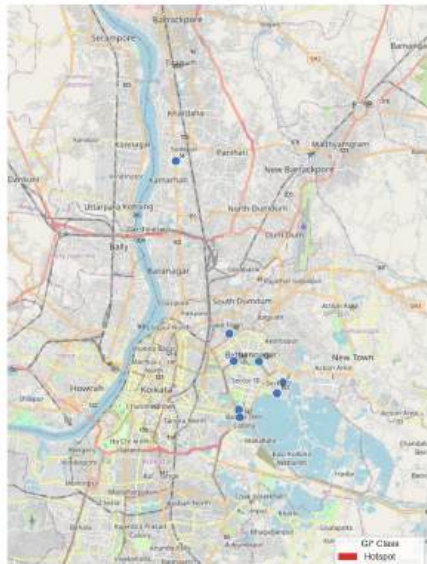


Comparison of Hotspot Maps

**GI* Hotspots/Coldspots
Durga Puja 2024 36° Day period**



**GI* Hotspots/Coldspots
Diwali 2024 26° Night period**



Comparison of Hotspot Maps

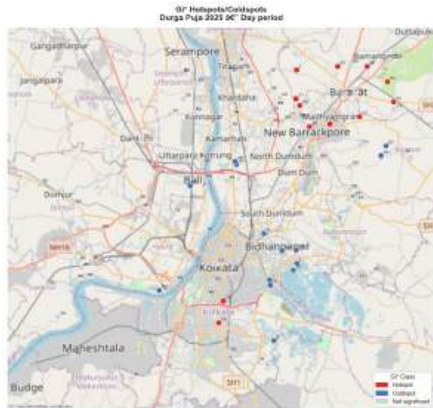


Figure: Daytime Hotspots

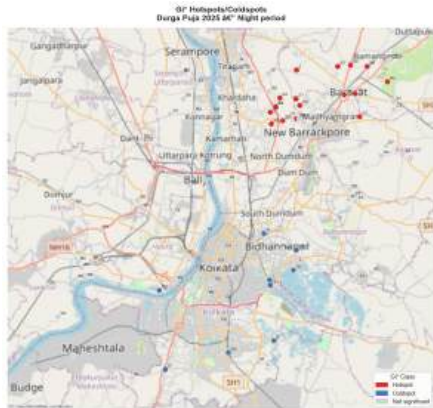


Figure: Nighttime Hotspots

Moran's I — Monthly (2022)

Month	Period	Moran's I	z	p (sim)	Decision
2*Mar	Day	—	—	0.001	Rejected
	Night	—	—	0.001	Rejected
2*Apr	Day	—	—	0.001	Rejected
	Night	—	—	0.001	Rejected
2*May	Day	0.3758	2.644	0.126	Not rejected
	Night	—	—	0.001	Rejected
2*Jun	Day	0.2987	2.694	0.122	Not rejected
	Night	-0.1429	-1.000	0.001	Rejected
2*Jul	Day	-0.1708	-0.175	0.354	Not rejected
	Night	-0.2341	-0.310	0.473	Not rejected
2*Aug	Day	0.1036	0.619	0.262	Not rejected
	Night	-0.2610	-0.388	0.493	Not rejected
2*Sep	Day	0.1435	0.803	0.204	Not rejected
	Night	-0.0104	0.257	0.334	Not rejected
2*Oct	Day	0.2867	1.276	0.119	Not rejected
	Night	-0.1741	-0.893	0.086	Rejected
2*Nov	Day	0.1419	0.857	0.212	Not rejected
	Night	0.1681	0.903	0.158	Not rejected
2*Dec	Day	0.0847	0.504	0.275	Not rejected
	Night	0.2269	1.353	0.104	Not rejected

Table: *

H_0 : No spatial autocorrelation in LAeq exceedance proportions. $\alpha = 0.10$, permutation-based p -value (999 permutations). [†] Moran's I and z unavailable for months with very few stations.

Moran's I — Monthly (2023)

Month	Period	Moran's I	z	p (sim)	Decision
2*Jan	Day	0.1385	0.926	0.181	Not rejected
	Night	-0.0826	-0.141	0.450	Not rejected
2*Feb	Day	0.1308	0.824	0.210	Not rejected
	Night	0.0219	0.344	0.306	Not rejected
2*Mar	Day	0.2907	1.529	0.079	Rejected
	Night	-0.0487	-0.021	0.405	Not rejected
2*Apr	Day	0.3543	1.834	0.032	Rejected
	Night	-0.0293	0.041	0.394	Not rejected
2*May	Day	0.2778	1.521	0.076	Rejected
	Night	0.1966	1.242	0.103	Not rejected
2*Jun	Day	0.4141	2.153	0.013	Rejected
	Night	0.0442	0.385	0.262	Not rejected
2*Jul	Day	0.4860	2.572	0.006	Rejected
	Night	0.1758	1.042	0.150	Not rejected
2*Aug	Day	0.5211	2.553	0.007	Rejected
	Night	0.1329	0.915	0.177	Not rejected
2*Sep	Day	0.5663	3.070	0.003	Rejected
	Night	0.2287	1.327	0.101	Not rejected
2*Oct	Day	0.6117	3.108	0.004	Rejected
	Night	0.1612	1.106	0.122	Not rejected
2*Nov	Day	0.5737	3.346	0.001	Rejected
	Night	0.1463	1.102	0.124	Not rejected
2*Dec	Day	0.3280	2.007	0.033	Rejected
	Night	0.1888	1.286	0.117	Not rejected

Table: *

Moran's I — Monthly (2024)

Month	Period	Moran's I	z	p (sim)	Decision
2*Jan	Day	0.3109	2.157	0.023	Rejected
	Night	0.0762	0.801	0.182	Not rejected
2*Feb	Day	0.1759	1.323	0.111	Not rejected
	Night	0.0143	0.286	0.286	Not rejected
2*Mar	Day	0.2430	1.644	0.051	Rejected
	Night	-0.1668	-0.859	0.172	Not rejected
2*Apr	Day	0.3037	2.183	0.021	Rejected
	Night	-0.1049	-0.480	0.381	Not rejected
2*May	Day	0.3414	2.217	0.015	Rejected
	Night	-0.1004	-0.426	0.381	Not rejected
2*Jun	Day	0.4772	3.175	0.002	Rejected
	Night	0.0445	0.533	0.240	Not rejected
2*Jul	Day	0.3653	2.458	0.007	Rejected
	Night	0.1184	0.927	0.144	Not rejected
2*Aug	Day	0.5282	3.521	0.001	Rejected
	Night	0.1598	1.284	0.109	Not rejected
2*Sep	Day	0.5033	3.193	0.004	Rejected
	Night	-0.0152	0.133	0.367	Not rejected
2*Oct	Day	0.5167	3.664	0.001	Rejected
	Night	0.0538	0.522	0.285	Not rejected
2*Nov	Day	0.5070	3.295	0.002	Rejected
	Night	-0.0013	0.194	0.370	Not rejected
2*Dec	Day	0.4619	3.343	0.002	Rejected
	Night	-0.0464	-0.157	0.479	Not rejected

Table: *

Moran's I — Monthly (2025)

Month	Period	Moran's I	z	p (sim)	Decision
2*Jan	Day	0.4279	2.953	0.003	Rejected
	Night	-0.0526	-0.159	0.477	Not rejected
2*Feb	Day	0.3720	2.975	0.003	Rejected
	Night	0.2766	2.210	0.017	Rejected
2*Mar	Day	0.2250	2.047	0.027	Rejected
	Night	0.1907	1.897	0.036	Rejected
2*Apr	Day	0.1626	2.421	0.009	Rejected
	Night	0.1570	2.369	0.021	Rejected
2*May	Day	0.1346	1.890	0.031	Rejected
	Night	0.2136	3.119	0.003	Rejected
2*Jun	Day	0.0670	1.067	0.153	Not rejected
	Night	-0.0045	0.065	0.434	Not rejected
2*Jul	Day	0.0106	0.300	0.377	Not rejected
	Night	0.0124	0.377	0.341	Not rejected
2*Aug	Day	0.0607	1.033	0.142	Not rejected
	Night	0.0604	0.924	0.179	Not rejected
2*Sep	Day	0.0839	1.327	0.098	Rejected
	Night	0.1579	2.129	0.026	Rejected
2*Oct	Day	0.0741	1.154	0.126	Not rejected
	Night	0.0798	1.258	0.122	Not rejected
2*Nov	Day	0.0466	0.809	0.205	Not rejected
	Night	0.0858	1.266	0.095	Rejected
2*Dec	Day	-0.0028	0.150	0.418	Not rejected
	Night	0.0970	1.573	0.072	Rejected

Table: *

Comparison of Hotspot Maps

GI* Hotspots/Coldspots
Oct 2023 @ 11th Day period

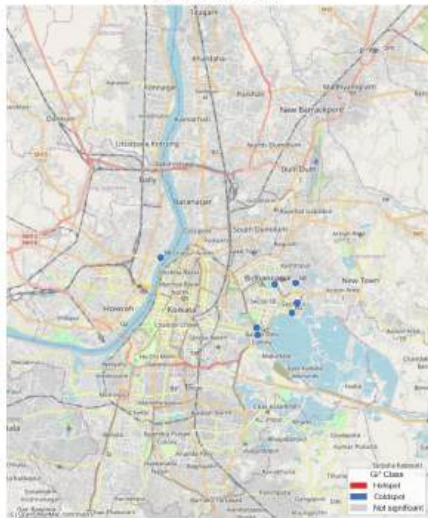


Figure: Daytime Hotspots

GI* Hotspots/Coldspots
Oct 2023 @ 11th Night period



Figure: Nighttime Hotspots

Comparison of Hotspot Maps

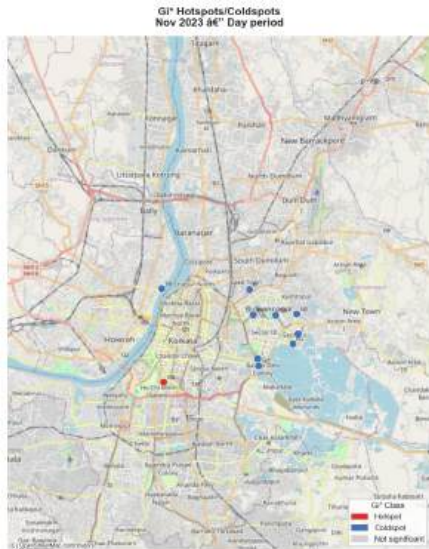


Figure: Daytime Hotspots



Figure: Nighttime Hotspots

Comparison of Hotspot Maps

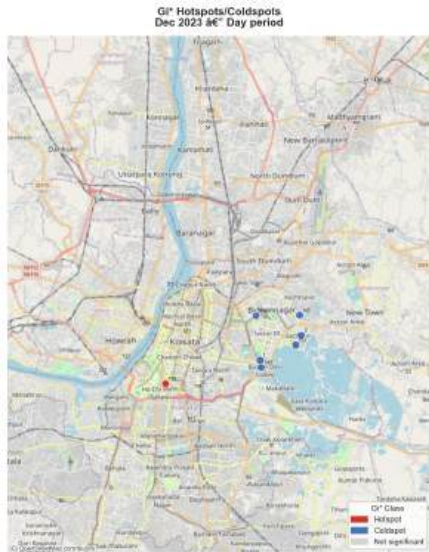


Figure: Daytime Hotspots

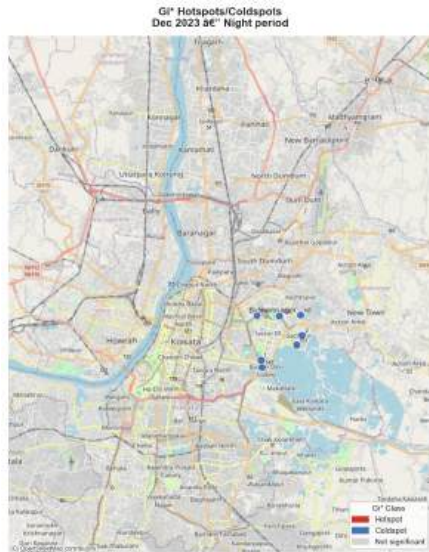
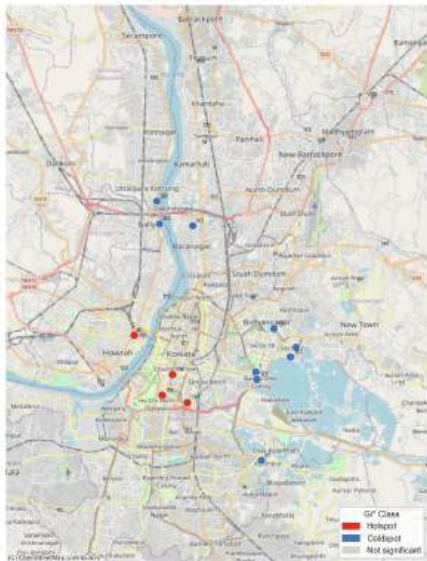


Figure: Nighttime Hotspots

Comparison of Hotspot Maps

GI* Hotspots/Coldspots
Nov 2024 56th Day period

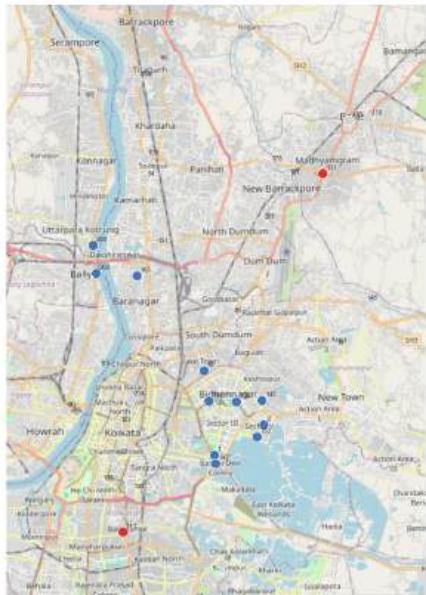


GI* Hotspots/Coldspots
Nov 2024 56th Night period

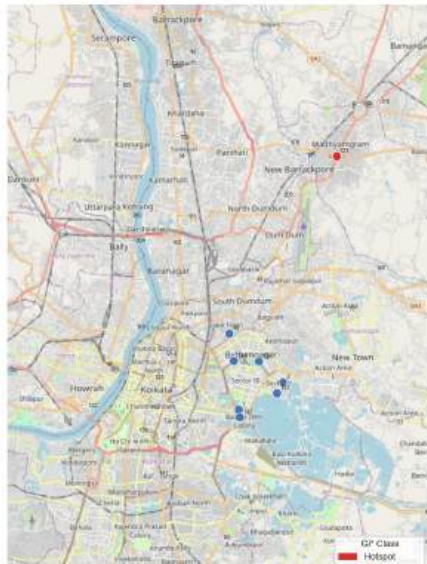


Comparison of Hotspot Maps

**GI* Hotspots/Coldspots
Oct 2024 a€" Day period**



**GI* Hotspots/Coldspots
Oct 2024 a€" Night period**



Comparison of Hotspot Maps



Figure: Daytime Hotspots



Figure: Nighttime Hotspots

Comparison of Hotspot Maps



Figure: Daytime Hotspots



Figure: Nighttime Hotspots

Comparison of Hotspot Maps



Figure: Daytime Hotspots



Figure: Nighttime Hotspots

Conclusions

- For checking global spatial autocorrelations, the null of the absence of the same, is rejected mostly for day periods. This behaviour seems to hold for both monthly and seasonal hotspots.
- As for festive seasonal hotspots, Durga Puja saw very few hotspots in 2023 and 2024, with some coldspots around the central region. In 2025, however, there were a lot more hotspots, mostly concentrated around the Madhyamgram-Barasat area.
- For Diwali, hotspots were noticed in again similar patterns as Durga Puja. Few were noticed in 2023, 2024 whereas 2025 saw quite a lot more hotspots, again concentrated around the Madhyamgram-Barasat area. Also, in earlier years, among the only hotspots being seen were from this area.
- As for Christmas-New year, similar trends were seen. In 2025, most hotspots were seen to be concentrated around the Ho-chi-Minh Sarani, and Ballygunge area.

Conclusions

- For monthly hotspots, similar trends were seen across the year. With very few hotspots observed during non-festive periods of the years.
- Hotspots, both monthly and festive seasonal, were mostly observed around the Ballygunge-Ho chi minh area and Madhyamgram-Barasat area.

Assessment of Environmental and Impulsive Noise Exposure in Kolkata Using Long-Term Monitoring Data

Abstract

- Environmental noise is a critical urban health challenge in densely populated cities like Kolkata.
- Traffic congestion and festive activities (fireworks, social gatherings) significantly escalate noise levels.
- This study evaluates long-term environmental and impulsive noise using high-resolution measurements (Jan 2023 – Jan 2025).
- Acoustic metrics analyzed: $LA_{eq,T}$, $LC_{eq,T}$, $LZ_{eq,T}$, $LA_{peak,T}$.
- Results indicate distinct temporal patterns and frequent exceedances of CPCB permissible limits.
- Festivals like Diwali and New Year exhibit substantial spikes in equivalent and peak noise levels.
- Data confirms the dominance of traffic and fireworks as primary acoustic sources.
- Findings emphasize the necessity of continuous monitoring for urban health and noise mitigation policy.

Introduction

- Kolkata's urban soundscape is dominated by vehicular traffic, construction, and cultural celebrations.
- Festive impulsive noise (Diwali, New Year) creates extreme acoustic exposure.
- Prolonged exposure is linked to hearing loss, sleep disturbance, and elevated stress.
- Vulnerable populations, particularly children and young adults, face heightened auditory risks.

Impulsive Noise in Urban Leisure Activities

- Impulsive noise consists of high-intensity, short-duration acoustic bursts.
- **Sources:** Fireworks, public celebrations, and sudden transport-related sounds.
- **Auditory Risks:** Immediate inner ear damage and risk underestimation by standard continuous metrics.
- Festive nighttime periods (Diwali, Kali Puja) exhibit critical acoustic spikes in Kolkata.

Environmental Noise Limits and Regulations

- **CPCB Permissible Limits (Commercial):**
 - Daytime (06:00–22:00): 65 dB(A)
 - Nighttime (22:00–06:00): 55 dB(A)
- **International Standards:** WHO Guidelines and EU Directive 2013/29/EU.
- **Objective:** Analyze environmental and impulsive noise patterns in Kolkata (Jan 2022 – Jan 2024).

Methodology: Study Area and Data Acquisition

- **Network:** Longitudinal data from 17 monitoring stations across Kolkata.
- **Selected Sites (SRSWOR):** Avidipta Housing Complex, Flora Fountain, and Presidency University.
- **Resolution:** Continuous 10-minute intervals (Jan 2022 – Dec 2023).
- **Scope:** High-resolution dataset capturing both continuous and impulsive noise characteristics.

Acoustic Indicators and Impulsive Noise

- **Primary Metrics:**

- $LA_{eq,T}$: Energy-equivalent A-weighted level.
- $LC_{eq,T}, LZ_{eq,T}$: Low-frequency and unfiltered energy indicators.
- $LA_{peak,T}$: Maximum instantaneous sound pressure level.

- **Impulsive Index (I_{idx}):**

$$I_{idx} = LA_{peak,T} - LA_{eq,T}$$

- Higher I_{idx} values identify high-energy, short-duration events (fireworks, traffic bursts).

Regulation of Impulsive Firecracker Noise

- **Indian Regulatory Standards (CPCB):**
 - Prohibited if noise exceeds 125 dB(AI) or 145 dB(C) peak at 4 meters.
- **Goal:** Mitigate hearing damage and environmental pollution during celebrations.
- These thresholds serve as reference points for identifying extreme impulsive events in the dataset.

Rule-Based Noise Classification

Noise events were classified using acoustic indicators and temporal activity conditions

Firecracker-Based Source

- High peak sound level:

$$L_{APeak,T} > 100$$

- Strong impulsive behaviour:

$$\text{Impulsiveness} > 30$$

- Nighttime activity period:

$$8 \text{ PM} - 4 \text{ AM}$$

Loud Music

- Strong low-frequency dominance:

$$L_{Ceq,T} - L_{Aeq,T} > 10$$

- Lower impulsiveness:

$$\text{Impulsiveness} < 15$$

Rule-Based Noise Classification (Contd.)

Traffic

- Elevated equivalent sound level:

$$L_{Aeq,T} > 60$$

- Limited low-frequency dominance:

$$L_{Ceq,T} - L_{Aeq,T} < 10$$

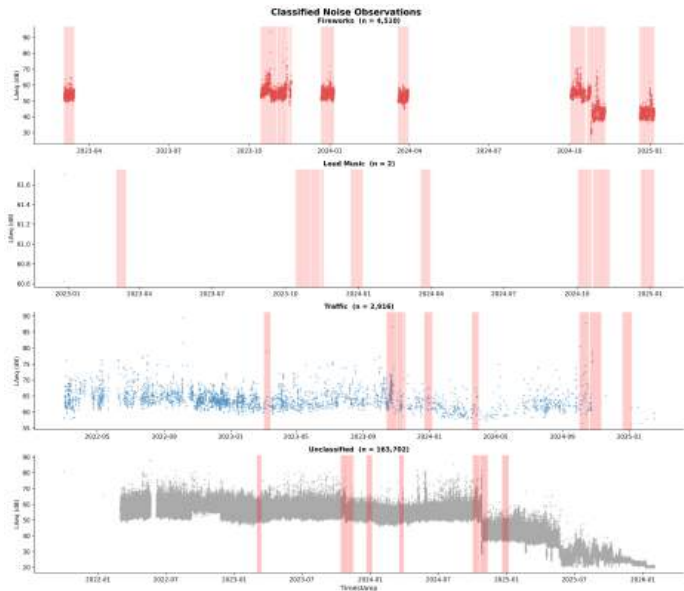
- Rush-hour activity periods:

9 AM – 1 PM, 5 PM – 8 PM

Unclassified

- Observations not satisfying predefined acoustic rules

Classified Noise Time Series



Interpretation of Classified Noise Observations

- **Fireworks:** Strong clustering during festivals (red shaded regions), particularly Diwali, Kali Puja, and New Year.
- **Traffic:** Persistent throughout the year, reflecting continuous urban vehicular activity.
- **Loud Music:** Relatively infrequent events within the monitored dataset.
- **Background Noise:** Majority of observations represent routine unclassified urban sounds.
- **Acoustic Intensity:** Festive periods show a marked increase in intensity and impulsive behavior.
- **Conclusion:** Festive celebrations significantly amplify urban acoustic exposure.

Overview of Acoustic Environment

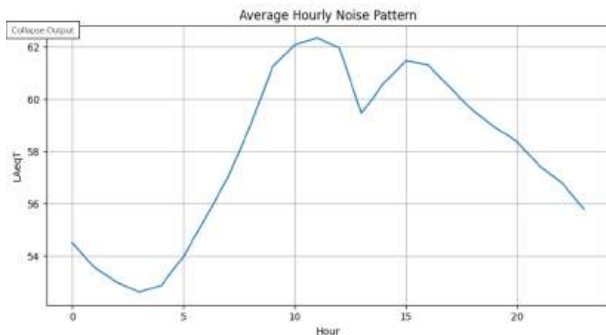
- Monitoring spanned Jan 2023 to Dec 2025, yielding 78,214 valid 10-minute observations.
- Results show persistently elevated noise levels across the study region.
- **Soundscape Drivers:** Traffic, human activity, commerce, and seasonal festivals.
- High temporal variability observed, particularly during festive periods.

Overall Acoustic Statistics

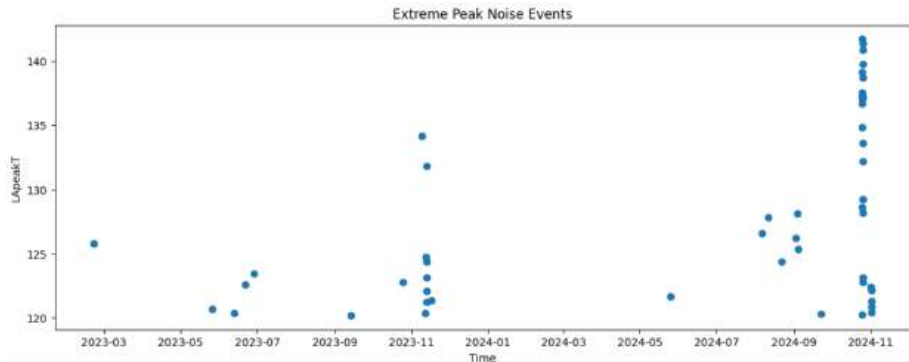
- $LA_{eq,T}$: Range 46.16–93.47 dB (Mean: 57.90 dB).
- $LA_{peak,T}$: Range 67.29–145.56 dB.
- **Frequency Metrics:** Mean $LC_{eq,T} = 65.86$ dB; Mean $LZ_{eq,T} = 70.58$ dB.
- **Percentile Levels:** $L_{10} = 63.44$ dB, $L_{50} = 57.69$ dB, $L_{90} = 52.38$ dB.

Average Hourly Noise Variation

- Noise levels peak during daytime activity, dip slightly around 14:00, and decline overnight.

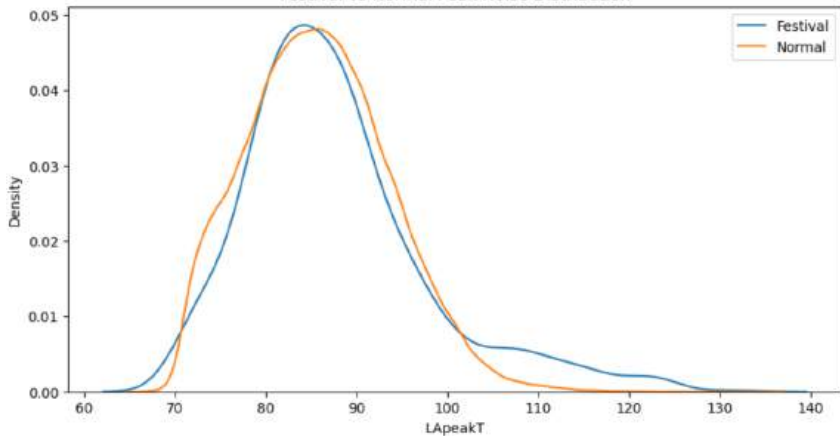


Extreme Peak Noise Events:



Note that in 2023, Diwali fell on November 12 and Durga Puja occurred from October 20 to 24 . In 2024, Diwali was observed on October 31, while Durga Puja took place from October 9 to 13.

Festival vs Normal Peak Noise Distribution



Kolmogorov–Smirnov (KS) Test Analysis

Hypotheses

$$H_0 : F_{\text{festival}}(x) = F_{\text{ordinary}}(x)$$

$$H_1 : F_{\text{festival}}(x) \neq F_{\text{ordinary}}(x)$$

Results

$$D = 0.0608$$

$$p = 0.0022$$

Interpretation

- Festival and ordinary noise distributions differ significantly
- Changes are mainly associated with:
 - Increased variability
 - Stronger impulsive acoustic events
 - Heavier upper-tail behaviour

Non-Parametric Analysis of Impulsive Noise

Average Impulsive Index

- Festival periods: 29.41 dB
- Ordinary periods: 27.90 dB

Mann–Whitney U Test

H_0 : Festival and ordinary periods have similar impulsive-index distributions

H_1 : Festival periods exhibit larger impulsive-index values

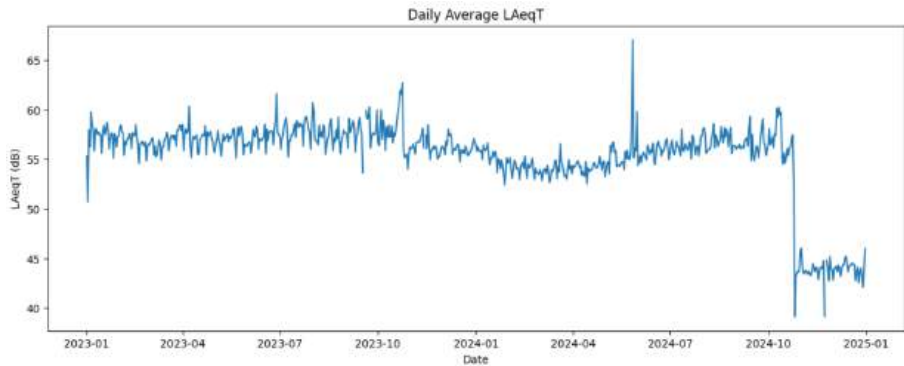
Results

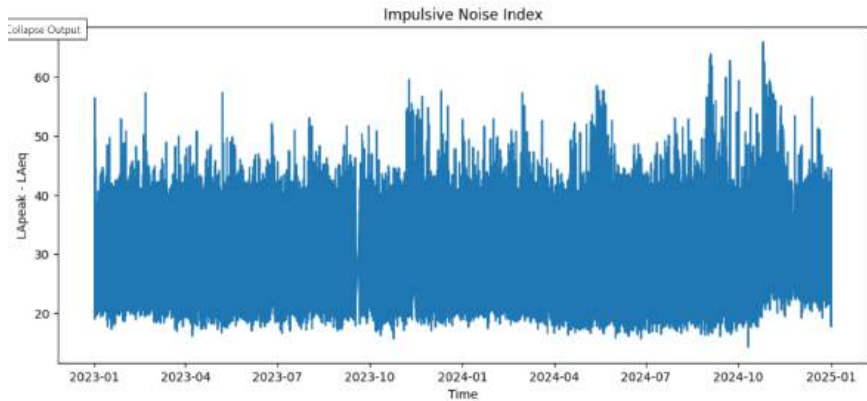
$$U = 38,827,721.5$$

$$p = 5.47 \times 10^{-6}$$

Conclusion

- Festival periods are associated with significantly stronger impulsive acoustic activity.





Conclusion

- Long-term monitoring confirms persistently high noise levels in Kolkata, frequently exceeding CPCB limits.
- Acoustic variability is primarily driven by seasonal festivals and urban traffic.
- Extreme impulsive events reached 145.56 dB, highlighting significant auditory risks.
- Statistical analysis confirms festivals exhibit significantly higher impulsive behavior than ordinary periods.
- **Policy Focus:** Prioritize continuous monitoring and stricter enforcement of festive noise regulations.

Challenge 1

- **Problem:** Reporting of 313 daily cycles and 313 annual cycles is difficult and visually uninformative.
- **Possible Approaches:**
 - ① Cluster similar cycles and report representative cluster centres.
 - ② Apply Functional Principal Component Analysis (FPCA) for dimension reduction.

Data Cleaning and Aggregation

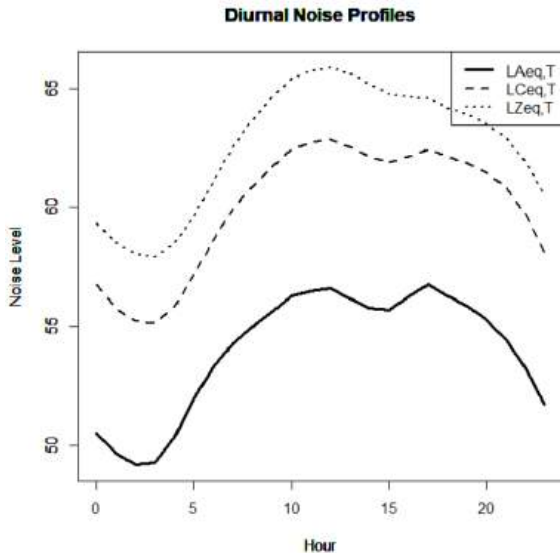
We have the following right now:

- 311 station-level .csv files.
- Approximately 36.6 million observations.
- Acoustic variables:
 - LAeq,T
 - LCeq,T
 - LZeq,T
 - LApeak,T
- Data collected from all districts of West Bengal.
- Timestamp-based temporal variables extracted:
 - Year
 - Month
 - Date
 - Time

Data Cleaning and Aggregation

- Observations prior to 2020 were removed due to anomalous timestamps.
- No missing values observed in principal acoustic variables.
- Raw high-frequency observations aggregated to hourly level.
- Hourly averages computed for:
 - $LA_{eq,T}$
 - $LC_{eq,T}$
 - $LZ_{eq,T}$
 - $LA_{peak,T}$

Overall Diurnal Noise Profiles

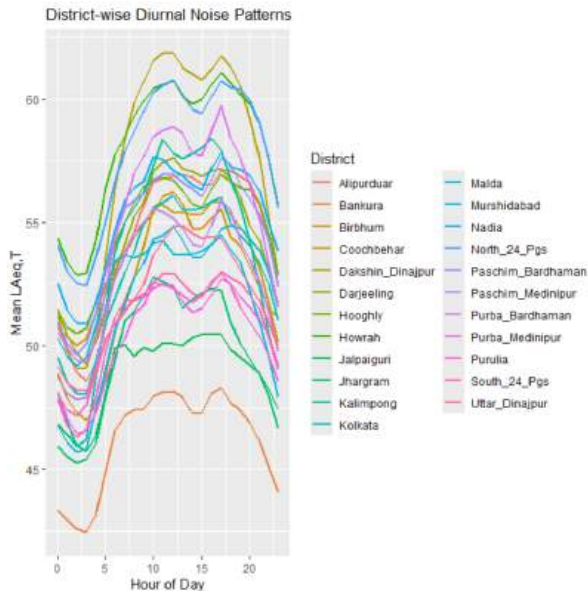


Overall Diurnal Pattern

- Clear daily temporal structure observed.
- Lower noise levels during late-night and early-morning hours.
- Rapid increase after morning activity begins.
- Elevated levels sustained during daytime and evening.
- Ordering observed:

$$LZeq, T > LCeq, T > LAeq, T$$

District-wise Diurnal Curves



District-wise Diurnal Variation

- All districts exhibited similar broad diurnal structure.
- Significant spatial heterogeneity observed.
- Some districts showed:
 - higher overall noise levels,
 - sharper morning peaks,
 - stronger evening activity.
- Motivated the need for dimension reduction methods.

Construction of Daily Functional Curves

Each daily trajectory represented as:

$$X_i(t), \quad t = 0, 1, \dots, 23$$

where:

- $X_i(t)$ denotes average noise level at hour t ,
- each row corresponds to one station-day curve.

Curves normalized using:

$$Z_i(t) = \frac{X_i(t) - \bar{X}_i}{s_i}$$

to focus on temporal shape rather than absolute loudness.

K-means Clustering of Daily Curves

- Representative sample of 10,000 curves used.
- Elbow method applied for cluster selection.
- Four-cluster solution selected.
- Cluster centres interpreted as representative daily acoustic patterns.

Statistical Procedure of K-means Clustering

- In this study, each daily noise cycle is treated as a 24-dimensional vector:

$$X_i = (x_{i1}, x_{i2}, \dots, x_{i24})$$

where:

- x_{i1} = noise level at hour 0,
 - x_{i2} = noise level at hour 1,
 - ...
 - x_{i24} = noise level at hour 23.
- Thus, every day becomes one point in a 24-dimensional space.
 - First, the number of clusters k is selected.
 - In the present analysis:

$$k = 4$$

- Initially, 4 random cluster centres are selected:

Statistical Procedure of K-means Clustering(Contd.)

- For every daily curve X_i , the distance from each cluster centre is computed.
- Euclidean distance was used:

$$d(X_i, \mu_j) = \sqrt{\sum_{h=1}^{24} (x_{ih} - \mu_{jh})^2}$$

- Each curve is assigned to the cluster whose centre is closest.
- After all curves are assigned, the mean curve inside each cluster is recomputed:

$$\mu_j = \frac{1}{n_j} \sum_{i \in C_j} X_i$$

where:

- C_j = set of curves in cluster j ,
- n_j = number of curves in cluster j .

Statistical Procedure of K-means Clustering(contd.)

- Therefore, each cluster centre becomes the average daily curve of that group.
- The procedure is repeated iteratively:
 - assign curves to nearest centres,
 - recompute centres,
 - assign again,
 - recompute again.
- The iterations continue until the clusters stop changing significantly.
- The algorithm minimizes the Within-Cluster Sum of Squares (WCSS):

$$\sum_{j=1}^k \sum_{i \in C_j} \|X_i - \mu_j\|^2$$

- WCSS measures total variability within clusters.
- Smaller WCSS indicates tighter and more homogeneous clusters.

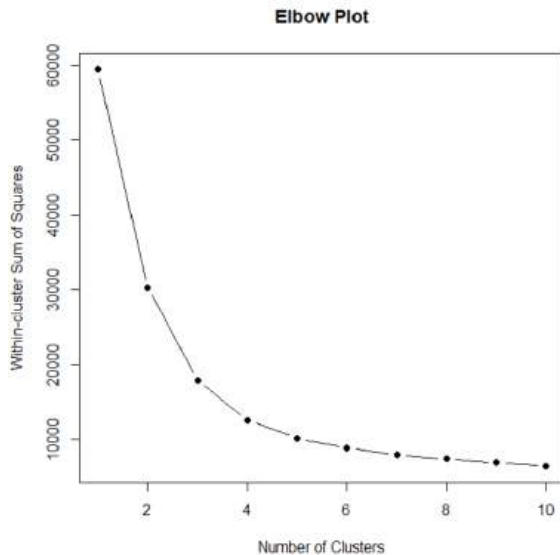
Selection of Number of Clusters: Elbow Method

- The K-means algorithm requires the number of clusters k to be chosen beforehand.
- To determine an appropriate value of k , the Elbow Method was used.
- K-means clustering was performed repeatedly for:

$$k = 1, 2, 3, \dots, 10$$

- For each value of k , the Within-Cluster Sum of Squares (WCSS) was computed.
- WCSS measures the total variability within clusters.
- As the number of clusters increases, WCSS always decreases because curves are grouped more precisely.
- However, after a certain point, the improvement becomes very small.

Elbow plot



Interpretation of Elbow Plot

- The WCSS values were plotted against the number of clusters k .
- Initially, the curve showed a sharp decline, indicating substantial improvement in clustering quality.
- Around:

$$k = 4$$

the rate of decrease in WCSS became much slower.

- This point appeared as a visible “elbow” in the graph.
- Therefore, $k = 4$ was selected because:
 - smaller values of k produced overly broad clusters,
 - larger values produced only marginal improvement,
 - four clusters provided a reasonable balance between compactness and interpretability.
- Hence, four representative daily noise patterns were finally retained.

Representative Cluster-centre Curves

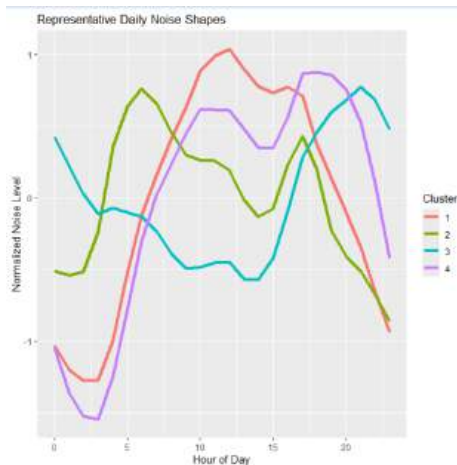


Figure: Representative Cluster-centre Curve

Functional Principal Component Analysis

- PCA applied to discretized daily curves.
- Provides low-dimensional representation of temporal variability.
- Variance explained:
 - PC1: 52.36%
 - PC1 + PC2: 62.39%
 - First 5 PCs: 76.5%
- Indicates strong low-dimensional temporal structure.

Functional Principal Component Analysis

- FPCA is based on the spectral decomposition of the covariance function.
- Covariance function:

$$C(s, t) = \frac{1}{n} \sum_{i=1}^n (X_i(s) - \mu(s))(X_i(t) - \mu(t))$$

- Functional principal components are obtained by solving the eigenfunction equation:

$$\int C(s, t) \phi_k(t) dt = \lambda_k \phi_k(s)$$

- Here:
 - $\phi_k(t)$: k -th eigenfunction,
 - λ_k : corresponding eigenvalue.

Functional Principal Component Analysis (Contd)

- Eigenfunctions represent dominant modes of variation in the daily curves.
- Eigenvalues measure the amount of variability explained by each mode.
- Larger eigenvalue:
 - more important component.
- Components are arranged in decreasing order:

$$\lambda_1 > \lambda_2 > \lambda_3 > \dots$$

- First few eigenfunctions summarize most temporal variability.

Functional Principal Component Analysis(Contd)

- After obtaining eigenfunctions, each daily curve is projected onto them.
- Principal component score:

$$\xi_{ik} = \int (X_i(t) - \mu(t))\phi_k(t) dt$$

- The score measures contribution of the k -th component to the i -th curve.
- Positive and negative scores correspond to different temporal behaviours.

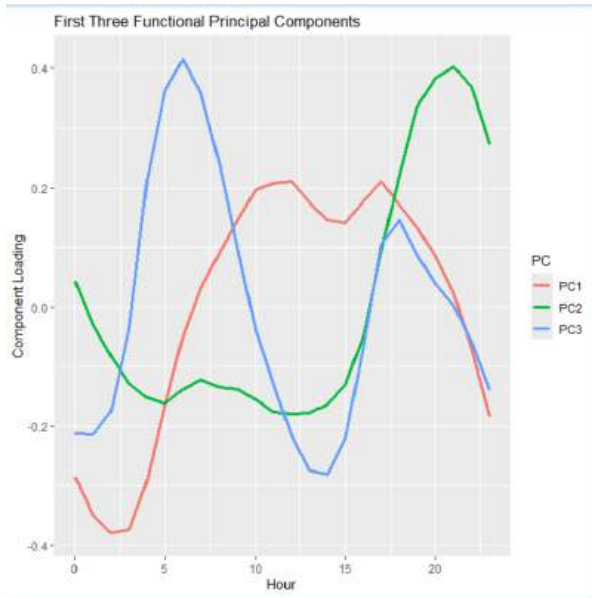
Functional Principal Component Analysis (Contd)

- Each daily curve can be approximated as:

$$X_i(t) = \mu(t) + \sum_{k=1}^K \xi_{ik} \phi_k(t)$$

- Thus:
 - mean curve captures overall behaviour,
 - eigenfunctions capture dominant variations,
 - scores quantify contribution for each day.
- Complex daily noise trajectories are therefore reduced to a few interpretable components.

First Three Functional Principal Components



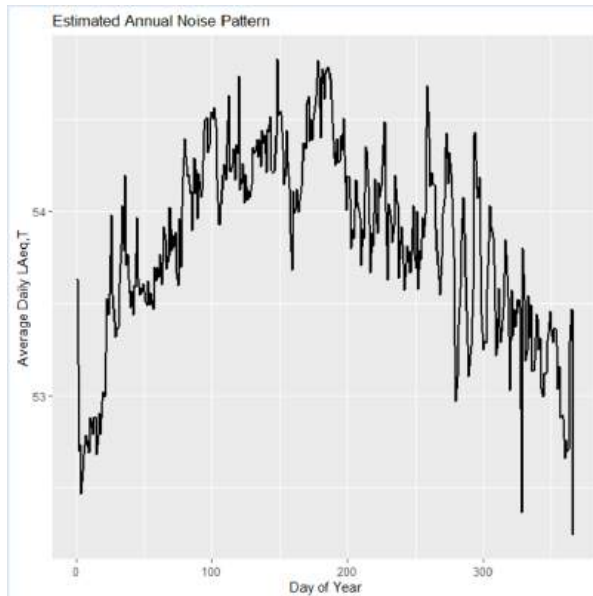
Interpretation of Principal Components

- PC1:
 - overall day-night contrast.
- PC2:
 - morning-dominant vs evening-dominant activity.
- PC3:
 - secondary fluctuations and nighttime irregularities.

Annual Pattern Estimation

- Daily average $LA_{eq,T}$ values computed.
- Annual trajectory estimated using day-of-year aggregation.
- LOESS smoothing applied to reveal long-term seasonal structure.
- Seasonal variation and episodic spikes observed.

Annual Noise Pattern



LOESS Smoothing Method

- LOESS (Locally Estimated Scatterplot Smoothing) is a non-parametric smoothing technique used to estimate an underlying trend from noisy data.
- Instead of fitting one global regression curve to the entire dataset, LOESS fits many small local regressions.
- For a given day t_0 :
 - observations near t_0 are selected,
 - nearby observations receive larger weights,
 - distant observations receive smaller weights,
 - a weighted local regression is fitted.
- The fitted value at t_0 becomes the smoothed estimate.
- This procedure is repeated for all calendar days to obtain a smooth annual trajectory.

LOESS Smoothing Method (Contd)

For each point t_0 , LOESS minimizes:

$$\min_{\beta_0, \beta_1} \sum_i w_i(t_0) (y_i - \beta_0 - \beta_1 t_i)^2$$

where:

- y_i : observed daily noise value,
- t_i : day-of-year,
- $w_i(t_0)$: weight assigned according to distance from t_0 .

Typical weight function:

$$w_i(t_0) = \left(1 - \left|\frac{t_i - t_0}{d}\right|^3\right)^3$$

Nearby observations obtain larger weights.

Smoothed Annual Trajectory

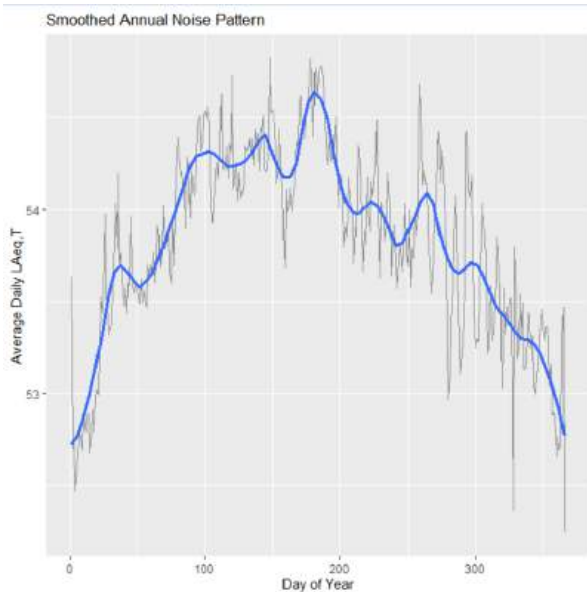


Figure: Smoothed Annual Trajectory

Conclusion

- Strong diurnal and seasonal variation observed in environmental noise.
- Significant spatial heterogeneity detected across districts.
- Clustering and FPCA successfully reduced extremely large collections of temporal curves.
- First few latent components captured most variability.
- Functional statistical techniques proved effective for large-scale environmental acoustic analysis.

Thank You!

Questions & Comments?